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# Benchmark Choice and Measured Test Size in Nested Tests of Cryptocurrency Return Predictability

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## Abstract

We ask whether daily cryptocurrency returns can be forecast out of sample well enough to survive transaction costs, and find that the answer turns almost entirely on which null the test states rather than on the data. Three assets, two horizons, and six forecasters are evaluated by purged and embargoed walk-forward cross-validation over 2020–2026, giving 18 machine-learning settings. The configuration in default applied use—a conditional model raced against a zero-return benchmark, corrected for nesting by Clark–West with a Newey–West bandwidth of  $h - 1$ —rejects 10–30% of the time at a nominal 5% on data resampled so that the features carry no predictive information. That is not a defect of the test. Clark and West [1] state its null as a *zero-mean* martingale difference, and these samples have a large drift, so the statistic is correctly rejecting a hypothesis the applied user did not intend to test. The mechanism is visible in the statistic: the estimators fit an unpenalised intercept, so the zero forecast is not their slopes-zero restriction, and the adjusted differential  $\hat{f}_t = 2(y_t - \hat{y}_t^b)(\hat{y}_t^m - \hat{y}_t^b)$  carries a drift term that is nonzero whatever the features do. Against the recursively estimated mean, which is the restriction these estimators do nest, measured size is 3–5% at  $h = 1$  and 7–8% at  $h = 7$  under a null that preserves the volatility clustering of the real series, including for early-stopped gradient boosting, for which we find no prior calibration evidence. The two benchmarks then support opposite readings of identical forecasts. Against zero, 5 of 18 settings reject and about 4 are expected from noise, so the count is uninformative. Against the recursive mean, 7 of 18 reject against about 1 expected—a real excess, of which two settings survive false-discovery control, none survives family-wise control, and the two survivors are near-identical linear models on one asset. That excess has no counterpart anywhere else in the study:  $R_{OS}^2$  is negative in 16 of 18 settings, the only significant sign-timing results ( $p \approx 0.001$  at  $h = 7$ ) arise from treating overlapping 7-day labels as independent and are  $p \approx 0.24$ – $0.27$  on non-overlapping subsamples, and the two settings whose net Sharpe intervals exclude zero lose most of it under a one-bar execution delay ( $0.91 \rightarrow 0.21$ ) while carrying  $\beta = 0.76$  and  $0.89$  to buy-and-hold with alpha  $t$ -statistics of 1.07 and 1.15. We also implement the Wild Clark–West statistic of Pincheira et al. [2], the published remedy for Clark–West’s long-horizon size distortion, and find it numerically indistinguishable from Clark–West here: it repairs a degeneracy, not a misspecified benchmark.

## 1 Introduction

Claims of accurate cryptocurrency price prediction are abundant and rarely survive scrutiny. The reported accuracy is usually traceable to one of four errors, each of which is well documented and each of which is easy to commit without noticing.

**Target leakage.** The predicted quantity is a function of contemporaneous or future information, so the model fits an identity rather than forecasting [3]. Leakage of this kind is now understood to be a leading cause of irreproducible findings in machine-learning-based science more broadly [4].

**Preprocessing leakage.** A scaler, imputer, or feature selector is fit on the full sample before splitting, so the test distribution informs training.

**Absent benchmarks.** Accuracy is reported on a price series rather than on returns. Prices are close to a martingale, so a naive random-walk forecast attains near-perfect  $R^2$ ; a model that appears to explain 99% of price variance may explain none of the return variance, which is the only component that can be traded [5].

**Silent multiple testing.** Many model, asset, and horizon combinations are tried and the best is reported without adjusting the threshold. In the closely analogous cross-sectional asset-pricing literature, Harvey et al. [6] argue that conventional significance levels are badly miscalibrated once the size of the collective search is acknowledged.

Preprocessing leakage is impossible by construction here, because standardisation is a pipeline step fit inside each fold. Target leakage is prevented by a test that perturbs future bars and asserts no feature at or before  $t$  changes. Multiple testing is reported over an explicit family, though not over the whole search. Benchmarking is the item this study initially got wrong.

The standard tool for comparing forecast accuracy, the Diebold–Mariano test [7], assumes the competing forecasts are non-nested. The comparison the return-predictability literature actually makes—a conditional model against a random walk—violates that assumption, and Clark and West [1, 8] give the correction. What the applied user must then check, and frequently does not, is which null the corrected statistic states. Clark and West [1] are explicit: theirs is that the series is a *zero-mean* martingale difference, with the zero forecast as benchmark. A regression fitted with an unpenalised intercept does not reduce to the zero forecast when its slopes are set to zero; it reduces to the training-window mean. Racing it against zero therefore tests no drift *and* no conditional predictability jointly, while the sentence such tests are used to support is about the features alone. Where the drift is large, as it is for these assets over this window, the two hypotheses are far apart. Section 4.8 derives the contaminating term, Section 4.9 measures what it does to the rejection rate, and Section 5.2 shows that repairing the benchmark changes which settings reject rather than how many.

**What is new here, and what is not.** The diagnosis above is not a discovery, and we state its provenance before reporting it. The zero-mean null is Clark and West’s own statement of their test [1]. The recursively estimated mean has been the standard out-of-sample benchmark in the equity-premium literature since Welch and Goyal [9] and Campbell and Thompson [10], and Moosa and Burns [11] show that the drift-versus-no-drift benchmark choice changes forecast-evaluation conclusions. Clark–West’s size distortion at long horizons was documented, and an asymptotically normal replacement—the Wild Clark–West statistic—proposed, by Pincheira et al. [2]; Magner and Hardy [12] apply that replacement to 13 cryptocurrencies over 2018–2022 against both a zero-forecast and a constant-forecast benchmark. A reader who knows that literature will find Section 4.8 unsurprising.

What we add is measurement rather than prescription. First, we calibrate instead of assuming: Section 4.9 measures the size of the entire applied pipeline—twelve technical features, purged walk-forward refits, ridge, elastic net, and an early-stopped gradient booster—under a null built by resampling the return series, and obtains 14–21% against the zero forecast and 3–5% against the recursive mean at  $h = 1$ . We found no prior calibration of Clark–West-family tests for adaptive machine-learning forecasters, where greedy split selection, subsampling, and a data-dependent number of trees might plausibly break the normal approximation. They do not appear to. Second, we implement the Wild Clark–West statistic and measure it in the same design, and report a negative result: in this signal-to-noise regime it is numerically indistinguishable from Clark–West, and it cannot in principle repair a benchmark that encodes the wrong hypothesis. Third, we report what these choices cost on a real study: four routine decisions—benchmark, HAC bandwidth, label overlap, and execution timing—each independently reversed a headline finding of an earlier version of this paper, and we give the before and after for all four.

That earlier version reported “five of 18 settings reject, against 0.9 expected from a search of that size.” The comparison assumed the test was correctly sized for the hypothesis we had in mind. It

was not, and the corrected expectation against that benchmark is about 4. We report the withdrawn claim explicitly because the error is easy to make, is invisible in the code, and is not detected by any test of the kind we had written.

## 2 Related work

**Return predictability and out-of-sample evaluation.** Welch and Goyal [9] showed that a long list of equity-premium predictors that perform well in sample fail to beat a simple historical mean out of sample, and Campbell and Thompson [10] formalised the out-of-sample  $R^2$  against that benchmark which we adopt in Section 4.6. The lesson generalises: in-sample fit is close to uninformative about forecast value, and the appropriate benchmark is a naive one estimated recursively rather than with hindsight.

**Predictive-accuracy tests for nested models.** Clark and McCracken [13] give the encompassing tests for nested forecast comparison and their non-standard limiting distributions; Clark and West [1] specialise the argument to the martingale-difference null, where the benchmark is the no-change forecast and the null asserts a *zero-mean* martingale difference; Clark and West [8] extend it to general nested pairs and report that with standard normal critical values the actual size is close to, and usually a little below, nominal. The degeneracy that makes the limit non-standard—under the null the two models coincide, so the core statistic has zero variance at population parameters—is diagnosed in West [14]. Pincheira et al. [2] remove it by multiplying the larger model’s error by an independent random variable with unit mean and small variance, giving the asymptotically normal Wild Clark–West statistic, which their simulations show is better sized at long horizons at some cost in power. Moosa and Burns [11] argue that whether the benchmark random walk carries a drift is an empirical question rather than a settled one, and that the answer changes which model wins. We use all of this rather than extend it; Section 4.9 measures how the resulting procedures behave in one concrete pipeline.

**Machine learning in asset pricing.** Gu et al. [15] conduct a large-scale comparison of machine-learning methods for equity returns and find measurable but small out-of-sample gains, concentrated in methods that shrink aggressively. Our model roster is chosen in that spirit: two regularised linear models and a shallow gradient booster (depth 3, learning rate 0.03, subsampling 0.8,  $\lambda = 1$ , early stopping), on the premise that an unregularised learner in this signal-to-noise regime fits noise. Those hyperparameters were fixed by hand before the grid was run and never tuned, which is a limitation and not a virtue: it keeps the search family small enough to enumerate, but it means no conclusion here concerns a model *family*.

**Backtest overfitting.** Bailey et al. [16] and Arnott et al. [17] document how repeated backtesting manufactures apparently significant strategies, and Bailey and López de Prado [18] propose the deflated Sharpe ratio, which raises the bar in proportion to the number of trials. White [19] and Sullivan et al. [20] address the same problem for technical trading rules through a bootstrap reality check. We adopt the deflated Sharpe ratio together with explicit control of the false discovery rate and the family-wise error rate over our grid.

**Cryptocurrency market efficiency.** The empirical record is mixed and appears to be time-varying. Urquhart [21] reports evidence of inefficiency in early Bitcoin returns; Nadarajah and Chu [22] find that a simple power transformation of the same returns satisfies efficiency tests, and Tiwari et al. [23] report that Bitcoin is largely efficient over most of their sample. Closest to our setting, Magner and Hardy [12] test 13 cryptocurrencies over 2018–2022 with autoregressive models and lagged Bitcoin returns, evaluated by Wild Clark–West against both the driftless random walk (a zero forecast) and the random walk with drift (a constant forecast), and report that their models beat the benchmark in a majority of exercises; Pincheira et al. [24] propose a further out-of-sample test aimed specifically at the random-walk benchmark. Reporting both benchmarks is therefore established practice in this literature, which is one reason we treat the benchmark point below as a measurement rather than a discovery. Much of this work evaluates statistical efficiency without asking whether any detected dependence survives transaction costs, which we do take up.

Table 1: Evaluated out-of-sample window per setting. Counts are forecasts, not trades; the trading rule of Section 4.5 samples every  $h$ -th bar.

| Asset | $h$ (days) | OOS forecasts | First      | Last       |
|-------|------------|---------------|------------|------------|
| BTC   | 1          | 2185          | 2020-07-23 | 2026-07-16 |
| BTC   | 7          | 2173          | 2020-07-29 | 2026-07-10 |
| ETH   | 1          | 2185          | 2020-07-23 | 2026-07-16 |
| ETH   | 7          | 2173          | 2020-07-29 | 2026-07-10 |
| SOL   | 1          | 1720          | 2021-10-31 | 2026-07-16 |
| SOL   | 7          | 1708          | 2021-11-06 | 2026-07-10 |

**Validation for dependent data.** Purged and embargoed cross-validation is due to López de Prado [25], and is the mechanism by which we prevent a training label whose realisation overlaps a test window from informing the fit.

### 3 Data

We use daily adjusted OHLCV bars from Yahoo Finance for the USD pairs of BTC, ETH, and SOL. BTC and ETH span 2019-01-01 to 2026-07-18 (2,756 bars each); SOL begins 2020-04-10 (2,291 bars). Cryptocurrency markets trade every calendar day, so the bar index is contiguous with no weekend or holiday gaps. We verified this rather than assuming it, and Section 4.3 shows that the split logic remains conservative under gaps regardless.

The evaluated out-of-sample window is shorter than the data window because the first fold consumes 504 bars of training history. Table 1 gives the resulting per-setting counts.

Asset selection is not survivorship-free. BTC, ETH, and SOL were chosen in knowledge of their survival, while many contemporaneous tokens did not survive. This biases the study toward finding profitable long exposure, which strengthens rather than weakens a negative result about *predictability*, but would invalidate any cross-sectional claim. Section 7 returns to it.

## 4 Method

### 4.1 Features

Let  $C_t, H_t, L_t, V_t$  denote close, high, low, and volume at bar  $t$ . Every feature is a function of information available at or before  $t$ , and all twelve are scale-free so that a single model can pool across assets trading at very different price levels. We use cumulative log returns over  $k \in \{1, 5, 10, 21\}$ , realised volatility over  $w \in \{10, 21\}$ , Wilder’s RSI over 14 bars recentred to roughly  $[-1, 1]$ , a price-normalised MACD histogram, two moving-average ratios, a mean normalised range, and a volume  $z$ -score:

$$r_t^{(k)} = \log \frac{C_t}{C_{t-k}}, \quad \sigma_t^{(w)} = \text{sd}\left(r_{t-w+1}^{(1)}, \dots, r_t^{(1)}\right), \quad (1)$$

$$\text{RSI}_t = 100 - \frac{100}{1 + \bar{U}_t / \bar{D}_t}, \quad \text{macd}_t = \frac{M_t - S_t}{C_t}, \quad (2)$$

$$\text{ma}_t = \log \frac{\text{SMA}_7(C)_t}{\text{SMA}_{21}(C)_t}, \quad \text{dist}_t = \log \frac{C_t}{\text{SMA}_{50}(C)_t}, \quad (3)$$

$$\text{range}_t = \frac{1}{14} \sum_{i=0}^{13} \frac{H_{t-i} - L_{t-i}}{C_{t-i}}, \quad z_t = \frac{\log(1 + V_t) - \mu_t^{(21)}}{s_t^{(21)}}, \quad (4)$$

where  $\bar{U}$  and  $\bar{D}$  are exponentially smoothed gains and losses with  $\alpha = 1/14$ , and  $M_t = \text{EMA}_{12}(C)_t - \text{EMA}_{26}(C)_t$  with  $S_t = \text{EMA}_9(M)_t$ . The longest warm-up is 50 bars; rows with any undefined feature are dropped.

The absence of look-ahead is enforced by a test rather than by inspection: future bars are perturbed and every feature value at or before  $t$  is asserted unchanged. This catches an accidental centred

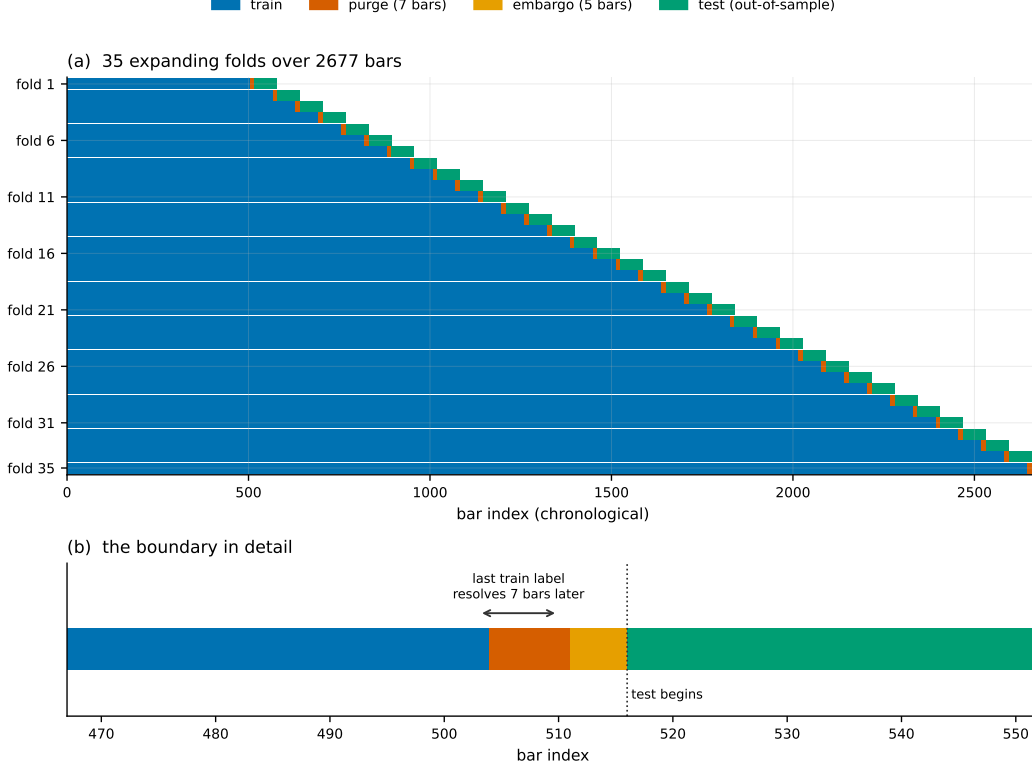


Figure 1: The validation design. Panel (a) shows all 35 folds on one timeline; panel (b) zooms on a single train/test boundary, where the purge and the embargo are individually visible. Equation (6) is what the gap enforces.

window, a sign error in a shift, or a non-causal smoother, none of which is reliably visible when reading code.

## 4.2 Prediction target

The label attached to a decision made at  $t$  is the return realised over  $(t, t + h]$ :

$$y_t^{(h)} = \log \frac{C_{t+h}}{C_t}. \quad (5)$$

The final  $h$  rows have no label. We retain them with features and a missing target, because that is exactly the state a live forecaster occupies on the most recent bar, and we exclude them from fitting and scoring.

## 4.3 Purged, embargoed walk-forward validation

Ordinary  $k$ -fold cross-validation shuffles time and trains on the future. A plain chronological split is also insufficient: because  $y_t^{(h)}$  resolves at  $t + h$ , a training row close to the test block carries a label whose realisation overlaps the test window. Following López de Prado [25], we purge the  $h$  rows before each test block and impose a further embargo of  $e$  bars. Writing  $\mathcal{T}_k$  and  $\mathcal{V}_k$  for the train and test index sets of fold  $k$ , every fold satisfies

$$\max(\mathcal{T}_k) + h + e < \min(\mathcal{V}_k). \quad (6)$$

Test blocks tile forward without overlap. We use an expanding window with  $|\mathcal{T}| \geq 504$  bars,  $|\mathcal{V}| = 63$  bars, and  $e = 5$ , giving 35 folds for BTC and ETH and 28 for SOL. Figure 1 shows the resulting layout, drawn from the splits the study actually runs.

Purging is applied by row position, whereas the leak it prevents is a calendar one. Position-based purging remains valid under missing bars, and conservatively so: gaps only stretch the timeline, so  $h$  positions always span at least  $h$  calendar days. We assert this on a synthetic index with a quarter of its bars removed.

Two further controls are worth naming because they are commonly missed. Standardisation is a pipeline step fit inside each fold, so it never sees test rows. And the gradient booster purges its own early-stopping holdout: without a gap, the last  $h$  rows of its fitting slice carry labels realised inside that holdout, so early stopping would be tuned on partly observed outcomes. We apply the same  $h$ -bar purge one level down, and test it by corrupting exactly those rows and asserting the fitted model is unchanged.

#### 4.4 Forecasters

Three benchmarks and three machine-learning models are refit from scratch on every fold. The benchmarks are the random walk  $\hat{y}_t = 0$ , the training-window mean  $\hat{y}_t = \bar{y}_T$ , and an AR(1) regression of  $y^{(h)}$  on  $r^{(1)}$ . The machine-learning models are ridge ( $\alpha = 1$ ), elastic net ( $\alpha = 10^{-3}$ ,  $\rho = 0.5$ ), and XGBoost [26] with depth 3, learning rate 0.03, subsampling 0.8,  $\lambda = 1$ , and early stopping on the purged temporal holdout described above. The linear objectives are

$$\hat{\beta}_{\text{ridge}} = \arg \min_{\beta} \|y - X\beta\|_2^2 + \alpha \|\beta\|_2^2, \quad \hat{\beta}_{\text{EN}} = \arg \min_{\beta} \frac{1}{2n} \|y - X\beta\|_2^2 + \alpha \rho \|\beta\|_1 + \frac{\alpha(1 - \rho)}{2} \|\beta\|_2^2. \quad (7)$$

Both linear objectives are fitted with an unpenalised intercept on standardised features, and the gradient booster estimates its own base score from the training labels. Setting the slopes to zero therefore returns the training-window mean of the target, not zero: measured on BTC at  $h = 1$ , the ridge and elastic-net intercepts equal  $\bar{y}_T$  to machine precision. The forecaster these models nest is the training-window mean, and the zero-return benchmark is the further restriction that the intercept is also zero. Section 4.8 shows what that distinction does to the test.

#### 4.5 Trading rule and transaction costs

Forecasts become unit positions on a non-overlapping schedule: one decision every  $h$  bars, held to the next. Sampling every  $h$ -th bar prevents the same  $h$ -day return being counted  $h$  times. Costs are charged on turnover, so reversing a position pays for two sides:

$$w_t = \text{sign}(\hat{y}_t) \in \{-1, 0, +1\}, \quad \tau_t = |w_t - w_{t-1}|, \quad \pi_t = w_t \left( e^{y_t^{(h)}} - 1 \right) - \tau_t c, \quad (8)$$

with  $c = 17$  basis points per side (10 bp taker fee, 5 bp slippage, 2 bp half-spread). Equity compounds as  $E_T = \prod_{t \leq T} (1 + \pi_t)$  from unity.

Because decisions are taken every  $h$  bars, the Sharpe ratio annualises at  $365/h$  rather than at 365. Using the latter inflates a seven-day strategy's Sharpe by a factor of  $\sqrt{7} \approx 2.6$ .

The schedule admits  $h$  start offsets and nothing distinguishes them. We therefore run every strategy on all  $h$  and report the spread; at  $h = 7$  it reaches 1.40 Sharpe points on one setting (Section 5.6).

An always-long reference is computed on the same schedule with the same cost model, so that a forecaster whose only achievement is holding the asset can be recognised as such.

#### 4.6 Out-of-sample $R_{OS}^2$

We report the Campbell–Thompson form [10], whose denominator is the squared error of a genuine ex-ante benchmark forecast rather than of the realised mean of the evaluation window:

$$R_{OS}^2 = 1 - \frac{\sum_t (y_t - \hat{y}_t)^2}{\sum_t (y_t - \hat{y}_t^b)^2}, \quad (9)$$

with  $\hat{y}^b$  the recursively estimated historical mean. That benchmark scores exactly zero against itself, which serves as a wiring check on the metric. The alternative denominator  $\sum_t (y_t - \bar{y})^2$  uses a quantity unknowable in advance and flatters the benchmark; we compute it as a cross-check but do not quote it.

Note that this is a *different* benchmark from the one the predictive-accuracy tests of Sections 4.7 and 4.8 use, and reporting effect sizes against one reference while testing significance against another is not defensible. It also happens to be the easier reference here: the zero forecast has lower squared error than the recursive mean in all six asset–horizon cells, by  $R_{OS}^2$  of +0.0005 (BTC  $h=1$ ) up to +0.1023 (SOL  $h=7$ ), the recursive drift estimate on SOL being badly wrong after the early-sample run-up. Measured against the zero forecast instead, exactly one of 18 settings has positive  $R_{OS}^2$ : BTC elastic net at  $h = 1$ , at +0.0026. Table 4 quotes the recursive-mean column; the zero-forecast comparison is the stricter one and is the number to read alongside a  $p$ -value computed against the zero forecast.

#### 4.7 Diebold–Mariano and the nested-model problem

For loss differential  $d_t = L(e_t^m) - L(e_t^b)$  under squared-error loss, the Diebold–Mariano statistic [7] is  $DM = \bar{d}/\sqrt{\hat{V}/n}$  with  $\hat{V}$  a Newey–West long-run variance [27] truncated at  $h - 1$  Bartlett lags, appropriate because  $h$ -step forecast errors follow an  $MA(h - 1)$ :

$$\hat{V} = \hat{\gamma}_0 + 2 \sum_{k=1}^{h-1} \left(1 - \frac{k}{h}\right) \hat{\gamma}_k, \quad \hat{\gamma}_k = \frac{1}{n} \sum_{t=k+1}^n (d_t - \bar{d})(d_{t-k} - \bar{d}). \quad (10)$$

The  $1/n$  normalisation applies to every autocovariance including the lagged ones. This is deliberate: the Bartlett weights guarantee  $\hat{V} \geq 0$  only under that normalisation, and the more intuitive  $1/(n - k)$  can drive a weighted sum negative in small samples. We apply the Harvey–Leybourne–Newbold small-sample correction [28],  $DM^* = DM\sqrt{(n + 1 - 2h + h(h - 1)/n)/n}$ , against a  $t_{n-1}$  reference. A negative statistic favours the model, and the test is two-sided because the null is equal accuracy and a model can be significantly worse.

The  $MA(h - 1)$  argument justifies the bandwidth for the *forecast error* under the null. It does not carry over to  $\hat{f}_t$  once Equation (12) applies, because the drift component  $2\mathbb{E}[y](\hat{y}_t^m - \hat{y}_t^b)$  inherits the persistence of the forecast. At  $h = 1$  the bandwidth is zero lags, so no autocovariance enters at all, and four of the five settings that reject in Section 5.2 are at  $h = 1$ . We therefore report both  $h - 1$  and the Newey–West plug-in bandwidth  $\lfloor 4(n/100)^{2/9} \rfloor$ , which is 8 lags at  $n \approx 2,200$ . On this data the plug-in bandwidth yields *smaller*  $p$ -values—the adjusted differentials are mildly negatively autocorrelated, so the  $h - 1$  choice is the conservative one at  $h = 1$ —but the count of surviving settings changes with it. The two bandwidths turn out to be interchangeable as far as size goes: under the independent null of Section 4.9 they differ by at most 0.3 percentage points in every cell we measured. What separates them is the rejection count on the real data, and nothing in the data selects between them, so we report the sensitivity as a result rather than choosing.

#### 4.8 Clark–West

Under a nested null, Diebold–Mariano is undersized as a test of predictability: the larger model pays sample mean squared prediction error to estimate coefficients that are truly zero, and the test reads that cost as evidence for the benchmark. Clark and West [8] subtract the term explicitly. With  $\hat{y}^b$  the nested benchmark and  $\hat{y}^m$  the larger model,

$$\hat{f}_t = (y_t - \hat{y}_t^b)^2 - \left[(y_t - \hat{y}_t^m)^2 - (\hat{y}_t^b - \hat{y}_t^m)^2\right] = 2(y_t - \hat{y}_t^b)(\hat{y}_t^m - \hat{y}_t^b), \quad (11)$$

and  $\bar{f}/\sqrt{\hat{V}_f/n}$  is referred to a standard normal, one-sided. A *positive* statistic favours the model, the opposite of the Diebold–Mariano convention; we report both statistics with their  $p$ -values throughout so the sign is never inferred from significance alone.

**What the benchmark choice does to the hypothesis.** The right-hand identity in Equation (11) is the whole issue. Clark–West is a  $t$ -test of whether the model’s *deviation from the benchmark* covaries positively with the outcome’s deviation from it. With  $\hat{y}^b = 0$  the identity collapses to  $\hat{f}_t = 2y_t\hat{y}_t^m$ , whose expectation is

$$\mathbb{E}[\hat{f}_t] = 2(\mathbb{E}[y]\mathbb{E}[\hat{y}^m] + \text{Cov}(y, \hat{y}^m)). \quad (12)$$

Only the second term is predictability. The first is drift multiplied by the average forecast, and it is nonzero whenever the sample has a drift and the model is on average long—which, through the fitted intercept, it always is here. BTC’s realised drift over this sample is +0.00102 per day, 37% annualised. Measured across the 18 settings, that term accounts for 9% of the numerator at BTC  $h = 1$  but 33% for the booster at  $h = 1$  and 41–61% across BTC  $h = 7$ . The benchmark forecasters make the point most directly: the training-window mean, which uses no features at all, scores  $CW = +1.03$  against the zero forecast on BTC at  $h = 1$ .

None of this is a fault in the statistic. Clark and West [1] define the null they test as a zero-mean martingale difference, so a drift term entering Equation (12) is the test doing what it says. The fault is in the translation: an applied paper that races a feature model against zero and reports “the features predict returns” has silently added “and the mean return is zero” to its null. We therefore report Clark–West against both the zero forecast, because that is what much of the applied literature does, and against the recursively estimated mean, which is the actual slopes-zero restriction of these estimators and the benchmark already used for  $R_{OS}^2$  in Section 4.6. Reporting one without the other conflates two hypotheses. Magner and Hardy [12] report both for cryptocurrencies, and Moosa and Burns [11] argue the general case.

#### 4.9 Calibrating the test instead of assuming it

Whether a rejection count is surprising depends on the test’s size, which we measure rather than assume. Resampling the daily log-return series into a synthetic price path makes future returns independent of every past-information feature by construction, so the null “the features add no predictive value” is true, while the realised drift, the fat tails, the feature collinearity and persistence, the sample length, the purged walk-forward geometry, and the estimators themselves are all retained. Forecasts are genuinely estimated at every origin, so the estimation-noise term Clark–West exists to remove is present—which is what distinguishes this from a simulation in which the larger model’s forecast is exogenous noise. In that latter construction the population mean squared prediction errors are  $\sigma_y^2$  and  $\sigma_y^2 + \sigma_u^2$ , so the null of equal accuracy is false rather than true, no coefficient is estimated, and Clark–West centres for the algebraic reason that  $\mathbb{E}[2y_t u_t] = 0$  for any independent  $u$  however poor a forecast it is. Such a design measures neither size nor bias, and an earlier version of this paper used it.

Resampling admits three nulls, and the distinction between them turns out to matter more than any other choice in this section.

**Independent.** Returns are drawn iid. All dependence is destroyed, including the volatility clustering that daily crypto returns certainly have, so this measures size against a null cleaner than reality.

**Block.** Returns are drawn in 21-day blocks. Volatility clustering survives, but so does whatever genuine within-block predictability the real series contains, so the rejection rate is an *upper bound* on size rather than a measurement of it.

**Sign-flipped.** Blocks are drawn as above, then the sign of each deviation from the sample mean is randomised:  $r_t = \mu + s_t(r_t^* - \mu)$  with  $s_t = \pm 1$  independent. Because  $|r_t - \mu|$  is untouched bar for bar, the volatility dynamics survive intact; because  $s_t$  is independent noise,  $\mathbb{E}[r_t | \mathcal{F}_{t-1}] = \mu$  exactly, so no conditional mean predictability remains. This is the null the other two cannot isolate: realistic dependence together with a true no-predictability hypothesis.

Table 2 gives rejection rates at a nominal 5% over 2,000 replications per cell (600 for the booster), with Monte Carlo standard errors.

Four readings follow.

First, the configuration in standard use rejects three to six times as often as nominal, and the drift-only row locates the cause: a forecaster with no feature information at all, which merely predicts the training-window mean, is called significantly better than the zero forecast in 27–42% of samples that contain no predictability whatever.

Second, substituting the recursive mean repairs it. Size is 3.1–3.7% at  $h = 1$  and 5.7–5.9% at  $h = 7$  under the independent null—slightly conservative, consistent with Clark and West [8]’s own report

Table 2: Rejection rate at a nominal 5% under three resampling nulls, with Monte Carlo standard errors. 2,000 replications per cell; 600 for the booster, which is measured at  $h = 1$  only. Return pool: BTC. Sign-flipped is the null that has both realistic dependence and no conditional predictability; the block column is an upper bound on size, not a measurement. Wild Clark–West was computed on every replication and its rejection rate differs from Clark–West’s by at most 0.4 percentage points in every cell, so it is not tabulated separately.

| Statistic                     | Model       | $h = 1$    |            |       | $h = 7$    |            |       |
|-------------------------------|-------------|------------|------------|-------|------------|------------|-------|
|                               |             | indep.     | sign-flip  | block | indep.     | sign-flip  | block |
| CW vs. zero forecast          | ridge       | 12.0       | 14.1       | 53.2  | 22.1       | 28.6       | 25.7  |
| CW vs. zero forecast          | elastic net | 17.5       | 20.9       | 43.9  | 23.4       | 30.4       | 25.1  |
| CW vs. zero forecast          | GBM         | 16.8       | 20.0       | —     | —          | —          | —     |
| CW vs. recursive mean         | ridge       | <b>3.7</b> | <b>4.8</b> | 44.6  | <b>5.9</b> | <b>7.2</b> | 10.7  |
| CW vs. recursive mean         | elastic net | <b>3.1</b> | <b>3.5</b> | 30.6  | <b>5.7</b> | <b>6.8</b> | 9.5   |
| CW vs. recursive mean         | GBM         | <b>3.7</b> | <b>3.0</b> | —     | —          | —          | —     |
| CW vs. zero, drift-only model | —           | 27.5       | 34.4       | 25.2  | 36.8       | 41.9       | 33.3  |
| DM vs. zero, two-sided        | ridge       | 61.0       | 55.5       | 15.0  | 50.6       | 46.9       | 46.8  |

Monte Carlo standard errors:  $\leq 0.6$  pp for entries below 10%,  $\leq 1.1$  pp elsewhere, and  $\leq 1.6$  pp in the 600-replication booster cells.

that the normal approximation is a little undersized—and it does so for the early-stopped booster as readily as for ridge. Greedy split selection, subsampling and a data-dependent number of trees do not visibly break the approximation here. That is calibration evidence for this estimator under this null, not a theorem, and we found no prior measurement of it.

Third, the dependence question has a cleaner answer than the block column suggests. Under block resampling the corrected test rejects 30–45% of the time at  $h = 1$ , which the earlier version of this paper read as a size failure under dependence. It is not: under the sign-flipped null, which preserves the same volatility dynamics but removes conditional predictability by construction, size is 3.0–4.8% at  $h = 1$  and 6.8–7.2% at  $h = 7$ . Almost all of the block column is therefore genuine within-block predictability in the real return series that block resampling carries into the synthetic path, not distortion in the test. The corrected test is usable under realistic dependence, mildly oversized at  $h = 7$ .

Fourth, Wild Clark–West makes no difference. Its rejection rate matches Clark–West’s to within 0.4 percentage points in every cell above, including the  $h = 7$  cells it was designed for. Section 5.2 gives the mechanism.

**What effect would this design detect?** Size answers half the question; a study that reports a null owes the other half. We index the alternative by an effect size rather than a nuisance scale: the sign-flipped deviations are passed through an AR(1) filter,  $r_t - \mu = \rho(r_{t-1} - \mu) + \sqrt{1 - \rho^2} d_t$ , which holds the unconditional variance fixed and gives the optimal one-step predictor a population  $R_{OS}^2$  of exactly  $\rho^2$ . The one-bar return is in the feature set, so the estimators can find it.

| $\rho$                | 0.02   | 0.04   | 0.06   | 0.08   | 0.10   | 0.11   | 0.12   |
|-----------------------|--------|--------|--------|--------|--------|--------|--------|
| population $R_{OS}^2$ | 0.0004 | 0.0016 | 0.0036 | 0.0064 | 0.0100 | 0.0121 | 0.0144 |
| Power, ridge          | 5.2%   | 11.4%  | 24.0%  | 42.8%  | 66.2%  | 75.0%  | 83.6%  |
| Power, elastic net    | 4.8%   | 14.0%  | 31.4%  | 55.2%  | 75.8%  | 83.8%  | 90.4%  |

Interpolating the ridge curve, the minimum detectable effect at 80% power is  $\rho = 0.116$ , a population  $R_{OS}^2$  of 1.34%. That is a large effect by the standards of this literature and roughly four times the largest  $R_{OS}^2$  this study observes. Two things follow, and they cut in opposite directions. Our failure to find predictability is not informative about effects below about one percent of return variance, so we make no claim there. And an effect small enough to hide from this design is small enough that the 17 bp cost model above would consume it.

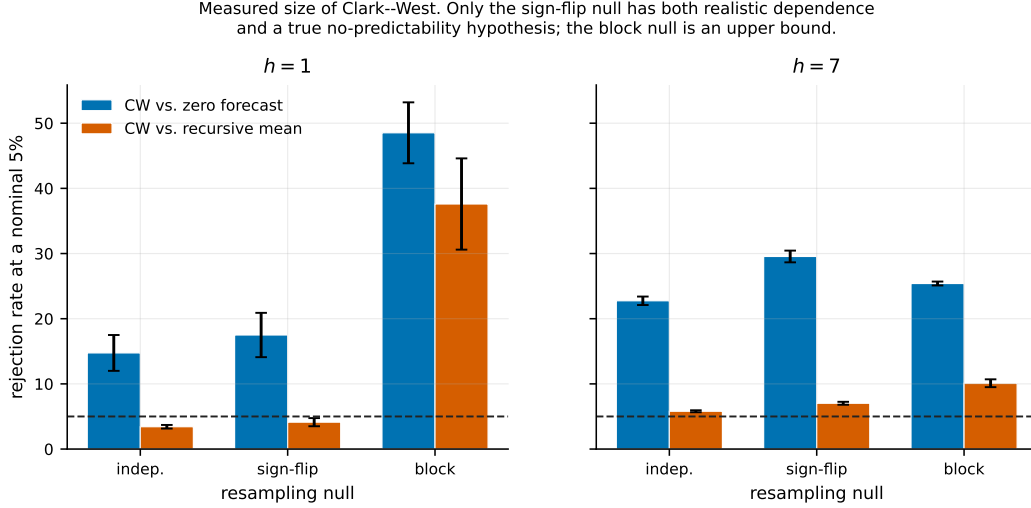


Figure 2: Measured rejection rate at a nominal 5% under the three resampling nulls. Bar height averages the estimators measured in each cell and the whisker spans them. Only the sign-flip null has both realistic dependence and a true no-predictability hypothesis; the block null is an upper bound on size, and the gap between the two is genuine within-block predictability rather than distortion in the test.

**Does the calibration transfer across assets?** Size measured on one return distribution need not hold for another, and the rejections in Section 5.2 are spread across all three assets. Repeating the sign-flipped experiment with each asset’s own return series as the pool, at 2,000 replications:

|                       | BTC            |                | ETH            |                | SOL            |                |
|-----------------------|----------------|----------------|----------------|----------------|----------------|----------------|
|                       | $h=1$          | $h=7$          | $h=1$          | $h=7$          | $h=1$          | $h=7$          |
| CW vs. zero forecast  | 14.1–20.9      | 28.6–30.4      | 9.8–12.1       | 19.8–20.4      | 9.7–11.9       | 22.9–23.9      |
| CW vs. recursive mean | <b>3.5–4.8</b> | <b>6.8–7.2</b> | <b>3.7–4.5</b> | <b>6.9–7.5</b> | <b>3.2–3.8</b> | <b>7.3–7.5</b> |

The recursive-mean calibration transfers: 3.2–4.8% at  $h = 1$  and 6.8–7.5% at  $h = 7$  across three assets whose daily volatilities differ by a factor of two. The zero-benchmark distortion does not transfer, and Equation (12) says it should not—it scales with drift relative to noise, so BTC, with the smallest volatility and a substantial drift, is the worst affected.

#### 4.10 Sign timing

A model can be poor at magnitude and useful at direction, which is all a sign strategy requires. Pesaran and Timmermann [29] test independence between the sign of the forecast and the sign of the outcome; their statistic  $S$  is asymptotically standard normal, positive for sign-timing skill and negative for a reliably wrong-way forecast. Appendix A states it. The assumption that matters here is that every variance term in it divides by  $n$ , so the  $n$  observations must be independent. Consecutive  $h$ -step forecasts built from overlapping windows are not, and Section 5.4 reports what that did to our own numbers. The statistic is undefined when the forecast never changes sign, which is the case for the always-long drift models, and we report it as missing rather than as a number.

#### 4.11 Sharpe ratio, deflation, and interval estimates

Sharpe ratios are noisy in short samples [30] and inflated by selection. We report the probabilistic Sharpe ratio of Bailey and López de Prado [18], the probability that the true Sharpe exceeds a threshold  $SR^*$  after correcting for sample length, skewness and kurtosis, and the deflated Sharpe ratio, which sets  $SR^*$  to the expected maximum over  $N$  trials so that the bar rises with the size of the search. Appendix A states both. Here  $N$  is the number of models raced within one asset-horizon,

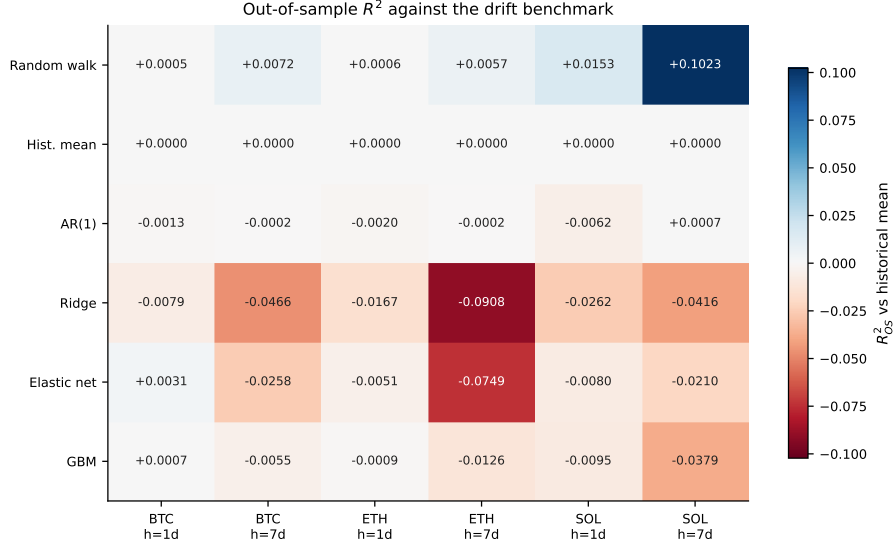


Figure 3: Out-of-sample  $R_{OS}^2$  against the drift benchmark. Red is worse than an ex-ante historical-mean forecast, blue better. The benchmark row is exactly zero by construction, which fixes the scale.

which is a floor on the true search and not an estimate of it, so the reported deflated Sharpe is an upper bound and is labelled as such. Confidence intervals come from a circular block bootstrap [31] with block length  $\lfloor n^{1/3} \rfloor$  and 500 resamples, reported as percentile intervals.

#### 4.12 Multiple testing

Eighteen settings tested by a correctly sized 5% test would produce  $0.05 \times 18 = 0.9$  rejections from noise alone. That is the right calculation only for a test with the size it claims. Averaging the sign-flipped column of Table 2 over the measured model–horizon combinations, the expected count is about 4 against the zero forecast and about 1 against the recursive mean. The zero-benchmark expectation is more than four times the nominal one, and the recursive-mean expectation is close to it. These, not 0.9, are the numbers a raw count must be read against, and which of them applies depends entirely on the benchmark. We report raw counts alongside two adjustments of the Clark–West  $p$ -values over the family of 18. Holm–Bonferroni [32] controls the family-wise error rate and Benjamini–Hochberg [33] the false discovery rate:

$$p_{(i)}^{\text{Holm}} = \max_{j \leq i} \min\{(m - j + 1) p_{(j)}, 1\}, \quad p_{(i)}^{\text{BH}} = \min_{j \geq i} \min\left\{\frac{m}{j} p_{(j)}, 1\right\}. \quad (13)$$

Benchmarks are excluded from the family: they are the null being tested against, not candidates in the search.

## 5 Results

### 5.1 Point accuracy

Out-of-sample  $R_{OS}^2$  is negative in 16 of 18 machine-learning settings, from  $-0.0908$  (ETH,  $h = 7$ , ridge) to  $+0.0031$  (BTC,  $h = 1$ , elastic net). Rank information coefficients lie between  $-0.080$  and  $+0.053$ . Figure 3 gives the full surface and Table 4 the underlying numbers.

That ridge attains  $\text{CW} = +2.62$  with  $p = 0.004$  on BTC at  $h = 1$  while its  $R_{OS}^2$  is  $-0.0079$  has two explanations, and only one of them was in the earlier version of this paper. Clark–West does ask about *population* mean squared prediction error, and a model can have a lower population error while its own parameter-estimation noise leaves the realised forecast worse than a constant. But the two numbers are also computed against different benchmarks—the  $p$ -value against zero, the  $R_{OS}^2$  against the recursive mean—and, per Equation (12) and Table 2, the  $p$ -value comes from a test that

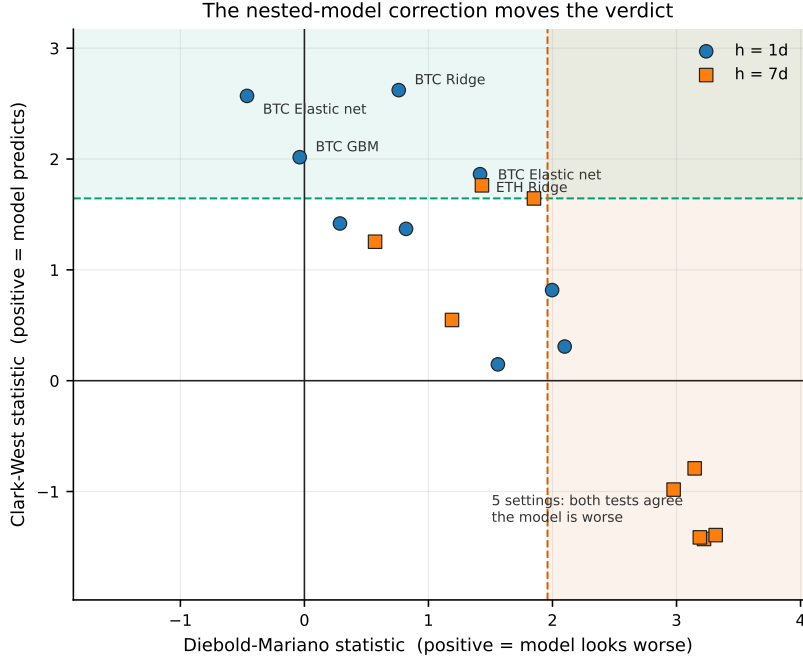


Figure 4: Every setting placed by its Diebold–Mariano statistic against its Clark–West statistic. The shaded vertical band marks where DM calls the model significantly worse; the horizontal band marks where Clark–West rejects no-predictability. Points falling in both are the settings on which the two tests give opposite readings of identical forecasts, which under nesting is expected rather than anomalous.

rejects 14% of the time when nothing is predictable. The gap is therefore not usable as evidence of a weak signal: both readings are available from the same forecasts, and the second explanation requires no signal at all.

## 5.2 Predictive accuracy against the random walk

Table 3 summarises. Under Diebold–Mariano the conclusion would be that machine learning is actively harmful here: no setting beats the random walk and seven lose to it significantly. Under Clark–West against the same zero benchmark, five settings reject at an uncorrected 5%. Both numbers derive from identical forecasts, and neither is the answer, because the second test rejects 14–30% of the time when nothing is predictable (Table 2).

Ordered by  $p$ -value the five are BTC ridge ( $p = 0.004$ ), BTC elastic net (0.005), BTC GBM (0.022), and ETH ridge (0.031), all at  $h = 1$ , together with BTC elastic net at  $h = 7$  (0.039). The sixth, BTC ridge at  $h = 7$ , misses at  $p = 0.05004$ ; a count that turns on four parts in  $10^5$  should not be described as five.

**The benchmark changes which settings reject, not how many.** Substituting the recursively estimated mean—the actual slopes-zero restriction of these estimators—moves the raw count from 5 of 18 to 7 of 18 at the same bandwidth. It does not move the same settings. BTC ridge and elastic net at  $h = 1$  survive ( $p = 0.004$  both). BTC at  $h = 7$  drops out ( $p = 0.086$  and  $0.061$ ). And SOL, which shows no signal at all against the zero benchmark, becomes the strongest rejecter: SOL GBM at  $h = 1$  goes from  $p = 0.207$  to  $0.031$ , SOL ridge at  $h = 7$  from  $0.923$  to  $0.022$ , SOL elastic net at  $h = 7$  from  $0.921$  to  $0.011$ . ETH at  $h = 7$  moves further against the models.

The earlier reading of this table—four of five rejections in BTC, consistent with a weak short-horizon effect in the most liquid asset—was an artifact of the benchmark. Under the corrected benchmark the geography reverses.

Table 3: Outcome of each test and correction across the 18 machine-learning settings. Direction is required, not merely significance: a Diebold–Mariano rejection is attributed to whichever side the statistic’s sign favours.

| Criterion                                               | Settings        |
|---------------------------------------------------------|-----------------|
| Diebold–Mariano, favours model                          | 0 / 18          |
| Diebold–Mariano, favours benchmark                      | 7 / 18          |
| Clark–West, uncorrected                                 | 5 / 18          |
| Clark–West, Benjamini–Hochberg                          | 2 / 18          |
| Clark–West, Holm–Bonferroni                             | 0 / 18          |
| Pesaran–Timmermann, sign skill                          | 0 / 18          |
| Pesaran–Timmermann, anti-predictive                     | 0 / 18          |
| Net Sharpe interval excludes zero                       | 2 / 18          |
| <i>Expected by chance at <math>\alpha = 0.05</math></i> | <i>0.9 / 18</i> |

Calibration then does something different to each count, and this is the result we would most like a reader to take away. Against the zero forecast, 5 of 18 is close to the  $\approx 4$  that the sign-flipped null produces when nothing is predictable, so the zero-benchmark rejections carry essentially no information. Against the recursive mean, 7 of 18 is well above the  $\approx 1$  expected, and that excess is not accounted for by size. The two benchmarks therefore support opposite readings of the same forecasts: under the default one the study has nothing to report, and under the one the estimators actually nest it has a small excess to report.

We are careful about how much that excess is worth. It does not survive family-wise control at any bandwidth, and only 2 of 18 survive false-discovery control at the  $h - 1$  bandwidth (5 at the plug-in bandwidth, with 0 and 1 surviving family-wise control respectively); since nothing in the data selects the bandwidth, the surviving set is not identified by this design. The two false-discovery survivors are BTC ridge and BTC elastic net at  $h = 1$ , which with  $\alpha = 10^{-3}$  are close to the same estimator on the same data, so they are one candidate finding rather than two. Every other measurement in this paper points the other way: those settings have negative  $R_{OS}^2$  against the same benchmark, no sign-timing skill, and no alpha once market exposure and a one-bar execution delay are accounted for.

**Wild Clark–West changes nothing here.** Pincheira et al. [2] restore asymptotic normality by multiplying the larger model’s error by  $\theta_t \sim \mathcal{N}(1, \phi^2)$  with  $\phi = c\widehat{\text{sd}}(\hat{e}^m)$  and  $c$  a small constant (Appendix A). Applied to these forecasts, the statistic is numerically almost identical to Clark–West: across all 36 setting–benchmark pairs the largest absolute difference is 0.0034 at  $c = 0.04$  and 0.0019 at  $c = 0.02$ , and no rejection decision at the 5% level changes under any  $c$  in the authors’ grid. The reason is a scale mismatch specific to this regime. The perturbation’s standard deviation is calibrated to the forecast error,  $\phi = c\widehat{\text{sd}}(\hat{e}^m)$ , whereas the core statistic’s own scale is set by the forecast difference  $\hat{y}^m - \hat{y}^b$ . Here the second is far smaller than the first, and the injected term contributes between 0.5% and 5.1% of the core statistic’s standard deviation at  $c = 0.02$  (median 1.5%), so the statistic barely moves. The single setting where the draw matters is BTC ridge at  $h = 7$ , whose  $p$ -value is 0.05004: there the 5% decision goes to the model in 43% of draws. Separately, and more fundamentally,  $\mathbb{E}[\theta_t] = 1$  leaves the drift term of Equation (12) untouched, so no version of the statistic can repair a benchmark that encodes the wrong hypothesis.

### 5.3 Multiple testing

Against the zero forecast, controlling the false discovery rate at 5% leaves BTC ridge and BTC elastic net at  $h = 1$ , both with  $p^{\text{BH}} = 0.046$ , marginally under the line; controlling the family-wise error rate leaves nothing, because the smallest raw  $p$ -value, 0.0044, does not clear the Holm threshold of  $0.05/18 = 0.0028$  (Figure 5). Those corrections are applied to  $p$ -values from a test whose size against this benchmark is 14–30% rather than 5%, so they do not control the error rates they name. This is not a technicality that cuts one way: the oversized test inflates the raw  $p$ -values’ evidential reading, while the multiplicity adjustment is calibrated to a nominal level the test does not have. No calibrated bound on the false-discovery rate over this family can be extracted from the zero-benchmark column.

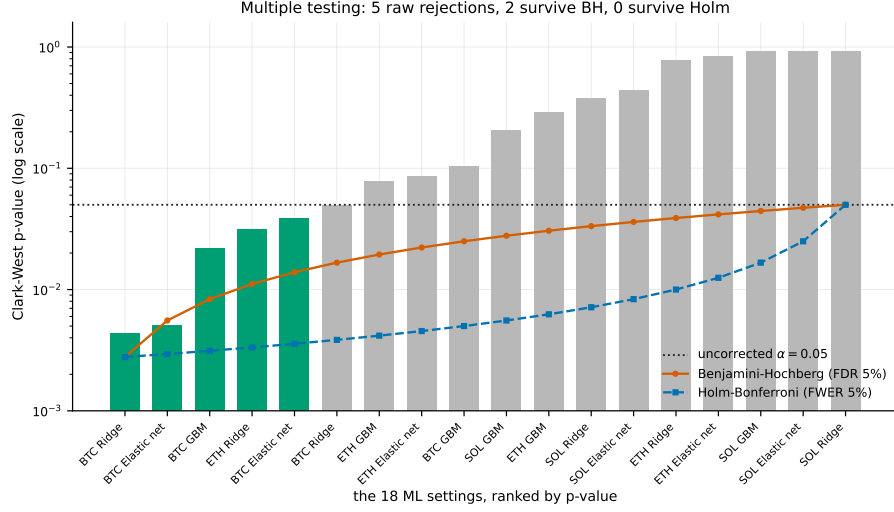


Figure 5: The 18 Clark–West  $p$ -values ranked against the Holm–Bonferroni and Benjamini–Hochberg thresholds, on a logarithmic scale because the entire decision occurs below  $p = 0.05$ . A bar must fall below a line to be rejected by that procedure. The figure’s reference line is the nominal one; the calibrated expectation against this benchmark is roughly 4 rejections rather than 0.9 (Table 2).

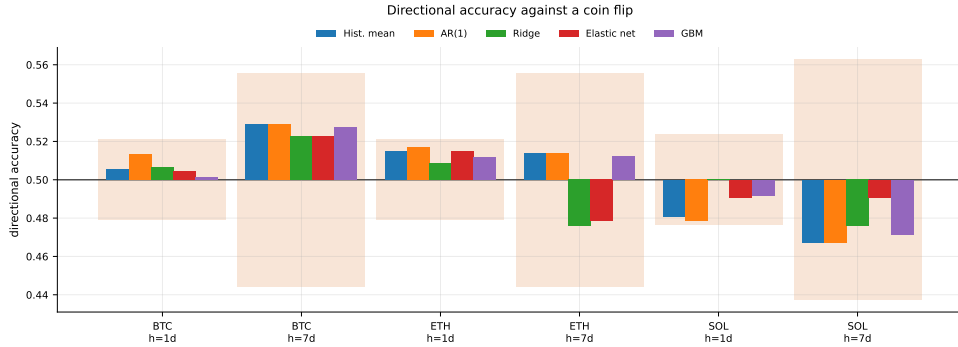


Figure 6: Directional accuracy against the band a fair coin would occupy. The band uses the effective sample size  $n/h$ : consecutive  $h$ -day forecasts are built from overlapping windows, and treating each as independent would shrink the band by  $\sqrt{h}$  and manufacture significance.

Against the recursive mean the adjustments are meaningful, because there the test is approximately correctly sized. The same two settings survive false-discovery control ( $p^{\text{BH}} = 0.035$ ), none survives family-wise control, and at the plug-in bandwidth the survivor counts are 5 and 1. So the calibrated reading of the whole family is: a raw excess of about six rejections over expectation, of which at most one distinct candidate survives false-discovery control and none survives family-wise control.

#### 5.4 Sign timing

Directional accuracy spans 0.471 to 0.528 and every setting lies inside its coin-flip band (Figure 6). No setting exhibits significant sign-timing skill.

The Pesaran–Timmermann variance terms all divide by  $n$ , so the statistic requires non-overlapping observations. An earlier version of this study fed it all  $n$  rows of an  $h$ -step forecast whose labels overlap  $h - 1$  times, which inflates the statistic by roughly  $\sqrt{h}$ . It was the same error Figure 6 avoids by drawing its band at  $n/h$ , so the table and the figure disagreed by a factor of  $\sqrt{7}$  at the longer

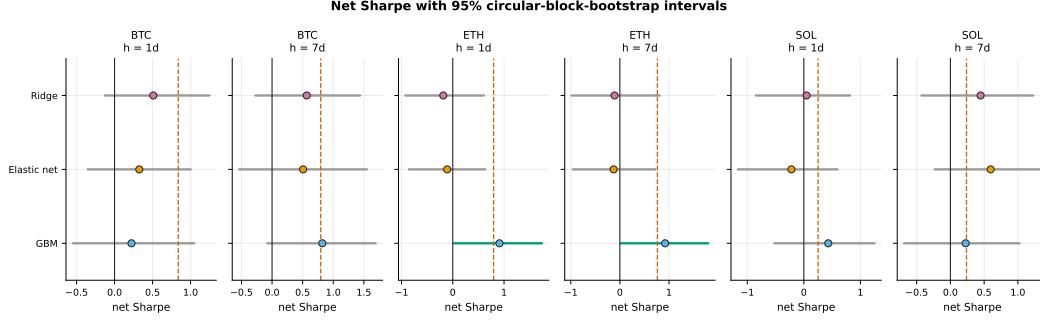


Figure 7: Net Sharpe with 95% circular-block-bootstrap intervals. The dashed line marks buy-and-hold on the same schedule and cost model. Intervals excluding zero are highlighted.

horizon. The pipeline now slices to  $y_{k::h}$  for each phase offset  $k$  and averages the statistic over all  $h$  of them, and Table 4 reports that version throughout. The four largest  $|S|$  at  $h = 7$ , before and after:

| Setting                  | $S$ (all $n$ ) | $p$   | $S$ (non-overlapping) | $p$   |
|--------------------------|----------------|-------|-----------------------|-------|
| ETH elastic net, $h = 7$ | −3.33          | 0.001 | −1.28                 | 0.242 |
| ETH ridge, $h = 7$       | −3.29          | 0.001 | −1.28                 | 0.265 |
| SOL ridge, $h = 7$       | −1.73          | 0.083 | −0.67                 | 0.604 |
| BTC ridge, $h = 7$       | +1.41          | 0.159 | +0.54                 | 0.596 |

The two ETH settings were previously reported as significantly *anti*-predictive at  $p \approx 0.001$ . They are not significant at any conventional level once the labels are non-overlapping, and neither is anything else. The conclusion—no sign-timing skill in either direction—is unchanged; the evidence previously offered for its most striking part was invalid.

## 5.5 Economic performance

Two of 18 settings have a net Sharpe whose interval excludes zero, both gradient boosting on ETH: 0.91 with interval  $[0.00, 1.74]$  at  $h = 1$ , and 0.92 with  $[0.02, 1.80]$  at  $h = 7$ . Both intervals begin essentially at zero. Neither setting appears among the two surviving Benjamini–Hochberg correction, and neither of those two has an interval excluding zero.

The statistical winners and the economic winners are therefore disjoint. Two unrelated groups of survivors drawn from one search is the signature of noise: if a genuine short-horizon effect existed and were tradeable, the settings that detect it and the settings that profit from it ought to overlap.

**Both economic winners are market exposure.** Regressing each strategy’s net return on buy-and-hold’s, on the same schedule and cost model, with a Newey–West standard error on the intercept:

| Setting                  | Sharpe    | % long | $\beta$ to buy-and-hold | $\alpha$ (ann.) | $t(\alpha)$  |
|--------------------------|-----------|--------|-------------------------|-----------------|--------------|
| ETH GBM, $h = 1$         | 0.91      | 94%    | 0.76                    | +0.236          | 1.07         |
| ETH GBM, $h = 7$         | 0.92      | 91%    | 0.89                    | +0.192          | 1.15         |
| BTC GBM, $h = 7$         | 0.82      | 90%    | 0.82                    | +0.097          | 0.66         |
| SOL elastic net, $h = 7$ | 0.60      | 56%    | 0.43                    | +0.499          | 1.23         |
| Historical mean (all)    | 0.23–0.84 | 100%   | 1.00                    | $\approx 0$     | $\leq  1.6 $ |

Neither of the two settings whose Sharpe interval excludes zero has an alpha distinguishable from zero once its market exposure is removed. An earlier version of this paper asserted the same thing in words, that the high-Sharpe results were long-only exposure; the regression is what makes it checkable.

**Execution is assumed at a price the signal has not yet been computed from.** Every feature at bar  $t$  uses the completed close  $C_t$ , and the backtest enters at that same  $C_t$ . Delaying entry by one bar, holding the forecasts fixed, moves the  $h = 1$  results substantially and the  $h = 7$  results hardly at all:

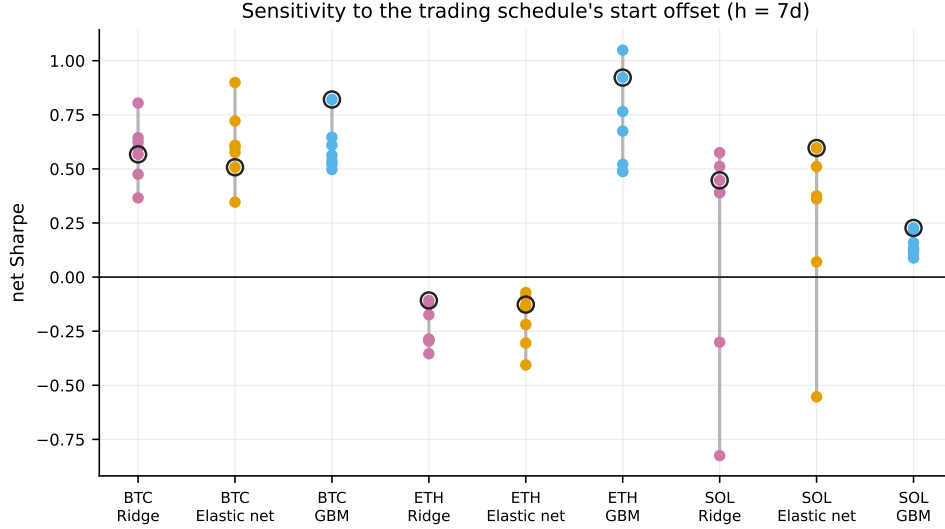


Figure 8: Net Sharpe on each of the  $h$  possible start offsets of the trading schedule at  $h = 7$ . The circled point is the offset the results table reports; nothing privileges it.

ETH GBM at  $h = 1$  falls from 0.91 to 0.21, BTC ridge at  $h = 1$  from 0.51 to  $-0.23$ , ETH elastic net at  $h = 1$  from  $-0.11$  to  $-0.61$ ; no  $h = 7$  setting moves by more than 0.20. The one-day results are therefore properties of the execution convention as much as of the forecasts, and the surviving economic result—ETH GBM at  $h = 7$ ,  $0.92 \rightarrow 0.98$ —is the one already shown above to be 91% long with  $\beta = 0.89$ .

**One cost scenario is not a cost model.** The 17 bp per side above is a single assumption. Sweeping it, with the forecasts held fixed, gives each setting a break-even cost—the per-side charge at which its net Sharpe reaches zero:

| Per-side cost                | 0 bp  | 17 bp | 25 bp | 40 bp | 100 bp |
|------------------------------|-------|-------|-------|-------|--------|
| Settings with net Sharpe > 0 | 16/18 | 13/18 | 11/18 | 9/18  | 6/18   |

The median break-even cost is 31 bp, roughly twice the assumed charge, so the sign of the economic result is not delicate at the margin. What the sweep exposes instead is the exposure finding again, from a different direction. The settings that survive the highest costs are the ones that barely trade: ETH GBM at  $h = 1$  breaks even at 148 bp and is 94% long, and the three settings with no finite break-even below 200 bp are all near-permanently long. Turnover, not forecast quality, is what determines cost tolerance here, and a strategy whose cost tolerance comes from never changing its mind is buy-and-hold with extra steps.

Six of 18 settings out-Sharpe buy-and-hold on the point estimate, the weakest form of that claim given that every interval contains zero. Meanwhile the historical-mean forecaster attains Sharpe ratios of 0.84, 0.80, and 0.25 across the three assets while making exactly one position change. It predicts a positive drift, hence is always long, and its net returns are numerically identical to buy-and-hold asset by asset and horizon by horizon. We carry buy-and-hold as an explicit row in Table 5 so that this cannot be mistaken for model performance.

Maximum drawdowns range from  $-55\%$  to  $-99\%$  across the machine-learning settings. This is a property of the position sizing rather than of any model: unit leverage reversing direction on roughly 70% annualised volatility carries a large variance drag, so the geometric return sits well below the arithmetic one. Buy-and-hold itself draws down between 76% and 96% over the same window.

## 5.6 Robustness to the trading schedule

Sampling every  $h$ -th bar leaves  $h$  schedules and nothing privileges the one starting at the first bar. At  $h = 7$  the spread is substantial: SOL ridge reports a Sharpe of 0.45 drawn from a distribution running from  $-0.83$  to  $0.57$ , and SOL elastic net reports 0.60 from a range of  $[-0.55, 0.60]$ . A single-offset Sharpe at a multi-day horizon is therefore one draw from a distribution 1.4 Sharpe points wide, and the reported figure is in part a property of where sampling began.

## 6 Discussion

**The correction inherits whatever null the benchmark encodes.** Diebold–Mariano is the default in applied forecast comparison and it is the wrong default whenever the benchmark is nested. Swapping in Clark–West fixes the nesting and leaves the hypothesis alone. Against a zero-return benchmark, an estimator with a fitted intercept is tested for drift and conditional predictability jointly, and Equation (12) shows the drift entering the statistic directly. The measured consequence in this study is a rejection rate three to six times nominal under a null with no conditional predictability in it, and a rejecting set whose membership changes from BTC to SOL when the benchmark is repaired. The operational form of the point is a question a referee can ask in one line: does the larger model fit an intercept, and if so, against which of the two nulls is the reported  $p$ -value computed?

**A distributional fix does not repair a hypothesis problem.** Wild Clark–West [2] is the right tool for the problem it was built for—the CW core statistic is degenerate under the null, which is why its limit is non-standard—and it is well sized at long horizons in the authors’ own designs. It cannot help here, and Section 4.9 shows that it does not: the drift term in Equation (12) survives the  $\theta_t$  perturbation unchanged, because  $\mathbb{E}[\theta_t] = 1$ . Choosing the benchmark and choosing the reference distribution are separate decisions, and only the first one determines what is being tested.

**Calibrate the test rather than assuming it.** The distortion above is invisible in code review, survives a unit-test suite that checks every formula against hand-computed values, and is not detected by simulating a forecast that is independent noise—a design that measures neither size nor bias and that an earlier version of this work used. What detects it is a null built by resampling the data through the actual pipeline with actual estimated forecasts. That experiment costs minutes of compute. Our own experience also argues for running it at a serious replication count: at 400 replications the size estimates moved by up to three percentage points between seeds, which is enough to change how a reader reads them.

**Most apparent performance was market exposure.** The highest Sharpe ratios in the study belong to a forecaster that makes one position change, and the two settings whose intervals exclude zero carry  $\beta = 0.76$  and  $0.89$  to buy-and-hold with alpha  $t$ -statistics of 1.07 and 1.15. A one-bar execution delay removes most of the  $h = 1$  performance. Three cheap controls—a buy-and-hold row, an exposure regression, and a delayed-entry variant—separate signal from exposure and from timing convention.

**What is left after all of this is small, and we do not want to overstate it in either direction.** Against the correct benchmark, with size calibrated under a dependence-preserving null, the raw rejection count is about six above expectation. That is a real statistical excess and we report it as one. It is also the weakest possible version of a finding: it does not survive family-wise control, the two settings that survive false-discovery control are two parametrisations of the same linear model on the same asset at the same horizon, the same settings are worse point forecasters than the benchmark they beat on the test, and nothing about them is tradeable once costs, execution timing, and market exposure are charged. In the other direction, the design detects a population  $R_{OS}^2$  of 1.34% only 80% of the time, so it is not powered to exclude an economically relevant effect either. The defensible summary is that the study finds one weak, uneconomic candidate and that the four apparently stronger findings of its earlier version were artifacts of the benchmark, the bandwidth, the label overlap, and the execution convention.

## 7 Limitations

Each item below is stated with what it would change if it were repaired, because a limitation that does not say that is a hedge.

**Limitations that bound the reported results.** The size calibration of Section 4.9 resamples all three assets’ return series, so size is established for those return distributions and not for other frequencies or estimators outside the three used here; the booster is calibrated at  $h = 1$  only. The sign-flipped null removes conditional mean predictability while preserving the volatility path, which is the dependence that matters most for daily returns, but it does not reproduce every feature of the real series—skewness dynamics and leverage effects, for instance, are destroyed with the sign. The minimum detectable effect is computed but large, so little here bounds predictability from above: a genuine  $R^2_{OS}$  of, say,  $+0.002$  would be entirely consistent with what we observe: the minimum detectable effect measured in Section 4.9 is a population  $R^2_{OS}$  of 1.34% at 80% power, so this design is blind to anything an order of magnitude smaller. The multiplicity family of 18 covers one statistic over one grid, and the hyperparameters were fixed by hand and never tuned, so every conclusion is about Ridge( $\alpha=1$ ), ElasticNet( $10^{-3}, 0.5$ ), and one booster configuration rather than about a model family; a tuned search would need its own multiplicity accounting, which would raise the bar rather than lower it. The sample end date was unpinned until this revision—the default configuration resolved it to the current day, so any run missing the committed cache silently re-dated the study—and the consequence was not hypothetical: one Monte Carlo pass was measured on a sample twelve bars longer than the manuscript’s and reported cell values differing by up to five percentage points. `StudyConfig.end` is now pinned to 2026-07-18 in the configuration rather than only by the presence of the cache.

**Limitations that bound generalisation.** Asset selection is not survivorship-free (Section 3); this biases towards finding profitable long exposure, which strengthens a negative result about predictability and would invalidate any cross-sectional claim. We use daily spot bars only, with no order book, funding rates, intraday structure, or cross-exchange information, and short-horizon predictability, if it exists, is more likely to live at frequencies this data cannot resolve—so the negative result should be read as being about daily bars specifically. Short positions are modelled as fully collateralised, returning  $-r$  with no borrow cost, margin, or liquidation, which is optimistic; correcting it would lower the reported Sharpe ratios and cannot raise them. Costs are a flat 17 bp per side that does not scale with size or volatility; we report a break-even sweep in Section 5.5 but not a queue-level execution model. Intervals are percentile bootstrap rather than BCa, adequate for a reject-or-not reading and mildly biased for a statistic as asymmetric as the Sharpe ratio. The deflated Sharpe ratio deflates for the six models raced within one asset-horizon—three of which are benchmarks—rather than for the whole grid or for the choices fixed before the grid was run, so it is an upper bound on the true corrected figure. Tightening it would not change a conclusion: the largest deflated Sharpe among the 18 settings is 0.847 (BTC GBM,  $h = 7$ ), already short of any conventional threshold.

## 8 Conclusion

Across 18 model, asset, and horizon combinations evaluated by purged walk-forward cross-validation from 2020 to 2026, the answer to whether daily cryptocurrency returns are predictable depends almost entirely on which null the test states. The configuration in standard use—Clark–West against a zero-return benchmark at an  $h - 1$  bandwidth—rejects 10–30% of the time on data constructed to contain no predictability, because a fitted intercept makes the zero forecast the wrong restriction and the sample drift enters the statistic directly. Its 5 of 18 rejections are therefore roughly what noise produces. Against the recursively estimated mean, which is the restriction these estimators do nest, the same statistic is approximately correctly sized—3–5% at  $h = 1$  and 7–8% at  $h = 7$  under a null that preserves volatility clustering, including for early-stopped gradient boosting—and it rejects in 7 of 18, against about 1 expected. That excess is the only positive result in the study, and it is a weak one: two settings survive false-discovery control and they are two parametrisations of one linear model on one asset, none survives family-wise control, the surviving set moves with the HAC bandwidth, and the same settings have negative  $R^2_{OS}$ , no sign-timing skill, and no alpha net of costs and market exposure. The only significant sign-timing results in the earlier version were an

overlapping-label artifact; the two settings with Sharpe intervals excluding zero carry  $\beta = 0.76$  and  $0.89$  to buy-and-hold with insignificant alpha, and the one-day one loses three quarters of its Sharpe to a one-bar execution delay.

We do not claim that no cryptocurrency signal exists at daily frequency; this design is not powered to support that claim and we make no attempt to. The transferable result is that four routine choices—which benchmark, which bandwidth, whether labels overlap, and when execution is assumed—determined every apparent finding in an earlier version of this study, and that all four are cheap to check. The benchmark is the one that has to be got right first, because it is the only one of the four that changes the hypothesis rather than the precision.

**Reproducibility.** All results, figures, and tables in this paper are generated by a single command from a public repository, and the tables reproduced here are emitted directly from the committed results file rather than transcribed. The test suite covers the leakage guarantees and checks the statistical formulas against hand-computed reference values. It did not, and could not, detect the benchmark misspecification of Section 4.8: every formula was implemented correctly and every test passed. The size calibration of Section 4.9, the corrected-benchmark re-test, the Wild Clark–West implementation, the non-overlapping sign tests, the exposure regressions, and the delayed-execution backtest are separate scripts under `audit/`, each reproducing the numbers quoted for it, together with a full record of which earlier claims they overturn. The per-observation forecasts on which every test in this paper is computed are committed as a file, so the inference can be re-analysed without re-running the models. Two cautions apply to anyone repeating the size experiment. It must be run with the sample end date pinned, for the reason given in Section 7; and it needs a serious replication count, because at 400 replications we observed cell-to-cell movement of up to three percentage points between random seeds.

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## A Statistical definitions

**Pesaran–Timmermann.** With  $P$  the hit rate,  $P_y$  and  $P_x$  the proportions of positive outcomes and forecasts, and  $P_* = P_y P_x + (1 - P_y)(1 - P_x)$ ,

$$S = \frac{P - P_*}{\sqrt{\widehat{\text{var}}(P) - \widehat{\text{var}}(P_*)}} \xrightarrow{d} \mathcal{N}(0, 1), \quad \widehat{\text{var}}(P) = \frac{P_*(1 - P_*)}{n}, \quad (14)$$

$$\widehat{\text{var}}(P_*) = \frac{(2P_y - 1)^2 P_x (1 - P_x)}{n} + \frac{(2P_x - 1)^2 P_y (1 - P_y)}{n} + \frac{4P_y P_x (1 - P_y)(1 - P_x)}{n^2}. \quad (15)$$

Every term divides by  $n$ , so  $n$  must count independent observations. Section 5.4 reports the statistic on non-overlapping subsamples for that reason.

**Probabilistic and deflated Sharpe ratio.** With skewness  $\gamma_3$  and kurtosis  $\gamma_4$ ,

$$\widehat{\text{PSR}}(\text{SR}^*) = \Phi \left[ \frac{(\widehat{\text{SR}} - \text{SR}^*)\sqrt{n-1}}{\sqrt{1 - \gamma_3 \widehat{\text{SR}} + \frac{\gamma_4 - 1}{4} \widehat{\text{SR}}^2}} \right], \quad (16)$$

and the deflated Sharpe ratio is  $\widehat{\text{PSR}}$  evaluated at  $\text{SR}^* = \mathbb{E}[\max_{i \leq N} \text{SR}_i]$ , approximated by

$$\mathbb{E} \left[ \max_{i \leq N} \text{SR}_i \right] \approx \sigma_{\text{SR}} \left[ (1 - \gamma) \Phi^{-1} \left( 1 - \frac{1}{N} \right) + \gamma \Phi^{-1} \left( 1 - \frac{1}{Ne} \right) \right], \quad (17)$$

with  $\gamma$  the Euler–Mascheroni constant.

**Wild Clark–West.** Let  $\hat{e}_t^b = y_t - \hat{y}_t^b$  and  $\hat{e}_t^m = y_t - \hat{y}_t^m$ . Pincheira et al. [2] replace the Clark–West core statistic  $\hat{f}_t = 2\hat{e}_t^b(\hat{e}_t^b - \hat{e}_t^m)$  with

$$\hat{f}_t^W = \hat{e}_t^b(\hat{e}_t^b - \theta_t \hat{e}_t^m), \quad \theta_t \stackrel{\text{iid}}{\sim} \mathcal{N}(1, \phi^2), \quad (18)$$

with  $\theta_t$  independent of the data. Under the null the two models coincide, so  $\mathbb{E}[\hat{f}_t^W] = \mathbb{E}[(\hat{e}_t^b)^2(1 - \theta_t)] = 0$  while the variance stays strictly positive, which restores the rank condition of West [14] and with it asymptotic normality. We set  $\phi = c\widehat{\text{sd}}(\hat{e}^m)$  with  $c \in \{0.01, 0.02, 0.04\}$  and average  $K = 2$  independent draws as the authors prescribe, dividing by  $\sqrt{\sum_i \sum_j \rho_{ij}}$  where  $\rho_{ij}$  is the sample correlation between the  $i$ th and  $j$ th core sequences.

## B Full results

Table 4 gives forecast accuracy and the predictive-accuracy tests for every model, asset, and horizon. Clark–West appears in two columns, against the zero forecast and against the recursively estimated mean; per Section 4.8 these are different hypotheses rather than a robustness check, and the second is the one whose size is calibrated in Table 2. Table 5 gives net-of-cost performance with bootstrap intervals, the phase spread, and both Sharpe corrections. Entries in bold are significant at the uncorrected 5% level; Section 5.2 gives the corrected counts, which are what should be read.

Table 4: Forecast accuracy and predictive-accuracy tests, all settings. Each test is reported as statistic ( $p$ ). DM negative favours the model, CW positive favours the model, PT positive indicates sign-timing skill.

| Model             | RMSE   | $R^2_{OS}$ | Dir. acc. | Rank IC | DM ( $p$ )           | CW vs. 0 ( $p$ )     | CW vs. mean ( $p$ )  | PT ( $p$ )    |
|-------------------|--------|------------|-----------|---------|----------------------|----------------------|----------------------|---------------|
| <i>BTC, h = 1</i> |        |            |           |         |                      |                      |                      |               |
| Random walk       | 0.0301 | 0.0005     | —         | —       | —                    | —                    | +1.38 (0.084)        | —             |
| Historical mean   | 0.0301 | 0.0000     | 0.506     | 0.002   | +0.18 (0.860)        | +1.03 (0.152)        | —                    | —             |
| AR(1)             | 0.0301 | -0.0013    | 0.513     | 0.033   | +0.34 (0.737)        | +1.24 (0.107)        | +0.86 (0.194)        | +1.23 (0.217) |
| Ridge             | 0.0302 | -0.0079    | 0.507     | 0.049   | +0.76 (0.447)        | <b>+2.62 (0.004)</b> | <b>+2.66 (0.004)</b> | +0.54 (0.591) |
| Elastic net       | 0.0301 | 0.0031     | 0.504     | 0.046   | -0.46 (0.644)        | <b>+2.57 (0.005)</b> | <b>+2.67 (0.004)</b> | +0.23 (0.817) |
| GBM               | 0.0301 | 0.0007     | 0.501     | 0.019   | -0.04 (0.970)        | <b>+2.02 (0.022)</b> | <b>+1.75 (0.040)</b> | -0.39 (0.698) |
| <i>BTC, h = 7</i> |        |            |           |         |                      |                      |                      |               |
| Random walk       | 0.0804 | 0.0072     | —         | —       | —                    | —                    | <b>+1.80 (0.036)</b> | —             |
| Historical mean   | 0.0807 | 0.0000     | 0.529     | -0.018  | +0.40 (0.686)        | +1.01 (0.156)        | —                    | —             |
| AR(1)             | 0.0807 | -0.0002    | 0.529     | -0.023  | +0.42 (0.675)        | +1.00 (0.159)        | -0.36 (0.639)        | —             |
| Ridge             | 0.0825 | -0.0466    | 0.523     | 0.044   | +1.85 (0.064)        | +1.64 (0.050)        | +1.36 (0.086)        | +0.54 (0.596) |
| Elastic net       | 0.0817 | -0.0258    | 0.523     | 0.043   | +1.43 (0.152)        | <b>+1.76 (0.039)</b> | +1.54 (0.061)        | +0.47 (0.633) |
| GBM               | 0.0809 | -0.0055    | 0.527     | 0.032   | +0.57 (0.569)        | +1.25 (0.105)        | +1.02 (0.153)        | +0.28 (0.723) |
| <i>ETH, h = 1</i> |        |            |           |         |                      |                      |                      |               |
| Random walk       | 0.0407 | 0.0006     | —         | —       | —                    | —                    | +1.30 (0.097)        | —             |
| Historical mean   | 0.0408 | 0.0000     | 0.515     | 0.019   | +0.25 (0.806)        | +0.81 (0.210)        | —                    | —             |
| AR(1)             | 0.0408 | -0.0020    | 0.517     | 0.036   | +0.47 (0.639)        | +1.16 (0.124)        | +0.91 (0.181)        | +0.95 (0.345) |
| Ridge             | 0.0411 | -0.0167    | 0.508     | 0.053   | +1.42 (0.157)        | <b>+1.86 (0.031)</b> | <b>+1.76 (0.039)</b> | +0.52 (0.600) |
| Elastic net       | 0.0409 | -0.0051    | 0.515     | 0.040   | +0.82 (0.413)        | +1.37 (0.085)        | +1.25 (0.106)        | +1.00 (0.315) |
| GBM               | 0.0408 | -0.0009    | 0.512     | 0.021   | +0.29 (0.775)        | +1.42 (0.078)        | +1.08 (0.139)        | -0.30 (0.762) |
| <i>ETH, h = 7</i> |        |            |           |         |                      |                      |                      |               |
| Random walk       | 0.1073 | 0.0057     | —         | —       | —                    | —                    | <b>+1.66 (0.048)</b> | —             |
| Historical mean   | 0.1076 | 0.0000     | 0.514     | 0.039   | +0.39 (0.696)        | +0.88 (0.188)        | —                    | —             |
| AR(1)             | 0.1076 | -0.0002    | 0.514     | 0.034   | +0.41 (0.685)        | +0.87 (0.191)        | -0.31 (0.623)        | —             |
| Ridge             | 0.1124 | -0.0908    | 0.476     | -0.047  | <b>+3.15 (0.002)</b> | -0.79 (0.786)        | -1.63 (0.948)        | -1.27 (0.265) |
| Elastic net       | 0.1115 | -0.0749    | 0.479     | -0.052  | <b>+2.98 (0.003)</b> | -0.98 (0.837)        | -1.80 (0.964)        | -1.28 (0.242) |
| GBM               | 0.1083 | -0.0126    | 0.512     | -0.042  | +1.19 (0.234)        | +0.55 (0.292)        | -0.36 (0.640)        | +0.09 (0.403) |
| <i>SOL, h = 1</i> |        |            |           |         |                      |                      |                      |               |
| Random walk       | 0.0499 | 0.0153     | —         | —       | —                    | —                    | <b>+4.75 (0.000)</b> | —             |
| Historical mean   | 0.0502 | 0.0000     | 0.481     | -0.029  | <b>+3.00 (0.003)</b> | -1.23 (0.891)        | —                    | —             |
| AR(1)             | 0.0504 | -0.0062    | 0.478     | -0.007  | <b>+2.39 (0.017)</b> | -0.77 (0.779)        | -0.41 (0.658)        | -0.81 (0.418) |
| Ridge             | 0.0509 | -0.0262    | 0.500     | 0.010   | <b>+2.10 (0.036)</b> | +0.31 (0.379)        | +1.34 (0.091)        | +0.01 (0.996) |
| Elastic net       | 0.0504 | -0.0080    | 0.491     | 0.013   | +1.56 (0.119)        | +0.15 (0.441)        | +1.43 (0.077)        | -0.66 (0.506) |
| GBM               | 0.0505 | -0.0095    | 0.492     | 0.023   | <b>+2.00 (0.046)</b> | +0.82 (0.207)        | <b>+1.87 (0.031)</b> | +0.57 (0.569) |
| <i>SOL, h = 7</i> |        |            |           |         |                      |                      |                      |               |
| Random walk       | 0.1346 | 0.1023     | —         | —       | —                    | —                    | <b>+5.23 (0.000)</b> | —             |
| Historical mean   | 0.1420 | 0.0000     | 0.467     | -0.103  | <b>+3.63 (0.000)</b> | -1.74 (0.959)        | —                    | —             |
| AR(1)             | 0.1420 | 0.0007     | 0.467     | -0.099  | <b>+3.62 (0.000)</b> | -1.71 (0.957)        | +1.29 (0.099)        | —             |
| Ridge             | 0.1449 | -0.0416    | 0.476     | -0.039  | <b>+3.22 (0.001)</b> | -1.43 (0.923)        | <b>+2.00 (0.022)</b> | -0.67 (0.604) |
| Elastic net       | 0.1435 | -0.0210    | 0.491     | -0.034  | <b>+3.19 (0.001)</b> | -1.41 (0.921)        | <b>+2.28 (0.011)</b> | -0.19 (0.504) |
| GBM               | 0.1447 | -0.0379    | 0.471     | -0.081  | <b>+3.32 (0.001)</b> | -1.39 (0.918)        | +0.37 (0.355)        | -0.18 (0.680) |

Table 5: Net-of-cost performance. Sharpe ratios annualise at  $365/h$ . Phases gives the range of net Sharpe across all  $h$  start offsets of the trading schedule. DSR is an upper bound, as discussed in Section 7.

| Model             | Sharpe | 95% CI        | Phases         | Max DD | PSR  | DSR  |
|-------------------|--------|---------------|----------------|--------|------|------|
| <i>BTC, h = 1</i> |        |               |                |        |      |      |
| Random walk       | 0.00   | [0.00, 0.00]  | –              | 0.0%   | –    | –    |
| Historical mean   | 0.84   | [0.01, 1.63]  | –              | -76.6% | 0.98 | 0.87 |
| AR(1)             | 0.21   | [-0.52, 0.91] | –              | -82.7% | 0.70 | 0.34 |
| Ridge             | 0.51   | [-0.13, 1.25] | –              | -65.7% | 0.89 | 0.63 |
| Elastic net       | 0.33   | [-0.35, 1.00] | –              | -74.3% | 0.79 | 0.45 |
| GBM               | 0.22   | [-0.55, 1.05] | –              | -89.4% | 0.71 | 0.35 |
| Buy & hold        | 0.84   | [0.01, 1.63]  | –              | -76.6% | 0.98 | –    |
| <i>BTC, h = 7</i> |        |               |                |        |      |      |
| Random walk       | 0.00   | [0.00, 0.00]  | [0.00, 0.00]   | 0.0%   | –    | –    |
| Historical mean   | 0.80   | [-0.12, 1.63] | [0.78, 0.80]   | -75.9% | 0.98 | 0.83 |
| AR(1)             | 0.80   | [-0.12, 1.63] | [0.78, 0.80]   | -75.9% | 0.98 | 0.83 |
| Ridge             | 0.57   | [-0.27, 1.44] | [0.37, 0.80]   | -78.5% | 0.92 | 0.65 |
| Elastic net       | 0.51   | [-0.54, 1.55] | [0.35, 0.90]   | -86.1% | 0.90 | 0.60 |
| GBM               | 0.82   | [-0.08, 1.69] | [0.50, 0.82]   | -54.6% | 0.98 | 0.85 |
| Buy & hold        | 0.80   | [-0.12, 1.63] | –              | -75.9% | 0.98 | –    |
| <i>ETH, h = 1</i> |        |               |                |        |      |      |
| Random walk       | 0.00   | [0.00, 0.00]  | –              | 0.0%   | –    | –    |
| Historical mean   | 0.80   | [-0.04, 1.59] | –              | -79.4% | 0.97 | 0.64 |
| AR(1)             | -0.15  | [-0.89, 0.60] | –              | -93.7% | 0.36 | 0.02 |
| Ridge             | -0.18  | [-0.93, 0.61] | –              | -97.5% | 0.33 | 0.02 |
| Elastic net       | -0.11  | [-0.86, 0.63] | –              | -97.7% | 0.39 | 0.03 |
| GBM               | 0.91   | [0.02, 1.74]  | –              | -80.4% | 0.99 | 0.74 |
| Buy & hold        | 0.80   | [-0.04, 1.59] | –              | -79.4% | 0.97 | –    |
| <i>ETH, h = 7</i> |        |               |                |        |      |      |
| Random walk       | 0.00   | [0.00, 0.00]  | [0.00, 0.00]   | 0.0%   | –    | –    |
| Historical mean   | 0.76   | [-0.09, 1.56] | [0.71, 0.76]   | -77.3% | 0.97 | 0.62 |
| AR(1)             | 0.76   | [-0.09, 1.56] | [0.71, 0.76]   | -77.3% | 0.97 | 0.62 |
| Ridge             | -0.11  | [-0.98, 0.81] | [-0.35, -0.11] | -99.2% | 0.40 | 0.03 |
| Elastic net       | -0.13  | [-0.95, 0.72] | [-0.41, -0.07] | -98.9% | 0.38 | 0.03 |
| GBM               | 0.92   | [0.02, 1.80]  | [0.49, 1.05]   | -84.5% | 0.99 | 0.76 |
| Buy & hold        | 0.76   | [-0.09, 1.56] | –              | -77.3% | 0.97 | –    |
| <i>SOL, h = 1</i> |        |               |                |        |      |      |
| Random walk       | 0.00   | [0.00, 0.00]  | –              | 0.0%   | –    | –    |
| Historical mean   | 0.25   | [-0.68, 1.22] | –              | -96.3% | 0.71 | 0.43 |
| AR(1)             | -0.19  | [-0.99, 0.69] | –              | -97.5% | 0.34 | 0.13 |
| Ridge             | 0.05   | [-0.85, 0.82] | –              | -97.4% | 0.54 | 0.27 |
| Elastic net       | -0.22  | [-1.17, 0.60] | –              | -97.1% | 0.32 | 0.12 |
| GBM               | 0.43   | [-0.52, 1.25] | –              | -91.3% | 0.83 | 0.59 |
| Buy & hold        | 0.25   | [-0.68, 1.22] | –              | -96.3% | 0.71 | –    |
| <i>SOL, h = 7</i> |        |               |                |        |      |      |
| Random walk       | 0.00   | [0.00, 0.00]  | [0.00, 0.00]   | 0.0%   | –    | –    |
| Historical mean   | 0.24   | [-0.80, 1.19] | [0.20, 0.25]   | -96.2% | 0.70 | 0.47 |
| AR(1)             | 0.24   | [-0.80, 1.19] | [0.20, 0.25]   | -96.2% | 0.70 | 0.47 |
| Ridge             | 0.45   | [-0.43, 1.23] | [-0.83, 0.57]  | -89.7% | 0.84 | 0.65 |
| Elastic net       | 0.60   | [-0.23, 1.34] | [-0.55, 0.60]  | -83.9% | 0.91 | 0.77 |
| GBM               | 0.23   | [-0.69, 1.03] | [0.09, 0.23]   | -94.9% | 0.69 | 0.46 |
| Buy & hold        | 0.24   | [-0.80, 1.19] | –              | -96.2% | 0.70 | –    |

## C Equity curves

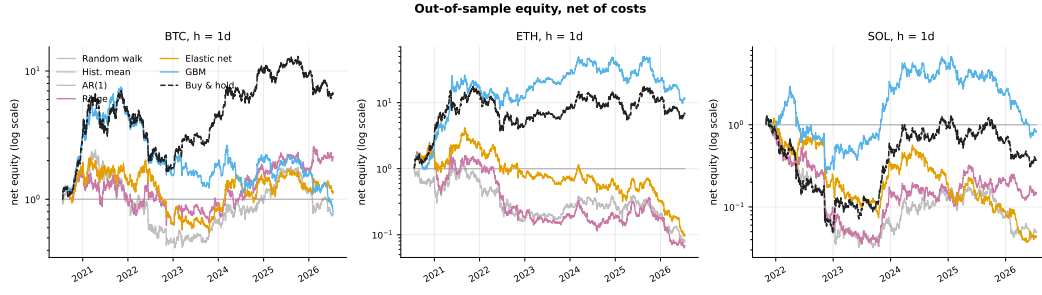


Figure 9: Out-of-sample equity net of costs at  $h = 1$ , against the always-long reference on the same schedule and cost model. Starting capital is unity and the vertical scale is logarithmic.